CE F230
Civil Engineering Materials

Special Concretes

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High Strength Concrete (HSC)

- Also known as ‘high performance concrete’
- High strength at either very early, or 28 days, or even later (> 80 MPa)
- Low permeability
- High durability
- High modulus of elasticity

High Strength Concrete (Ingredients)

- Type I (or Type III) Portland cement at a very high content (450 to 550 kg/m^3)
- Silica fume, generally 5 to 15% by mass of the total cementitious materials
- Sometimes, fly ash or ground granulated blast furnace slag
- Superplasticizer (5 to 15 liters/m^3 of concrete) → it can reduce the water content of about 45 to 75 kg/m^3 of concrete
- Good quality and well-graded aggregates

High Strength Concrete

- Low w/c ratio (< 0.35)
- Mix must be sufficiently workable
- Adequate compaction and curing
- To ensure good bond between the coarse aggregate particles and matrix, these particles should be approximately equi-dimensional
- Smaller maximum size of aggregate
- Fine aggregate should not be rounded and uniformly graded, but rather coarse; a fineness modulus between 2.8 - 3.2 is recommended

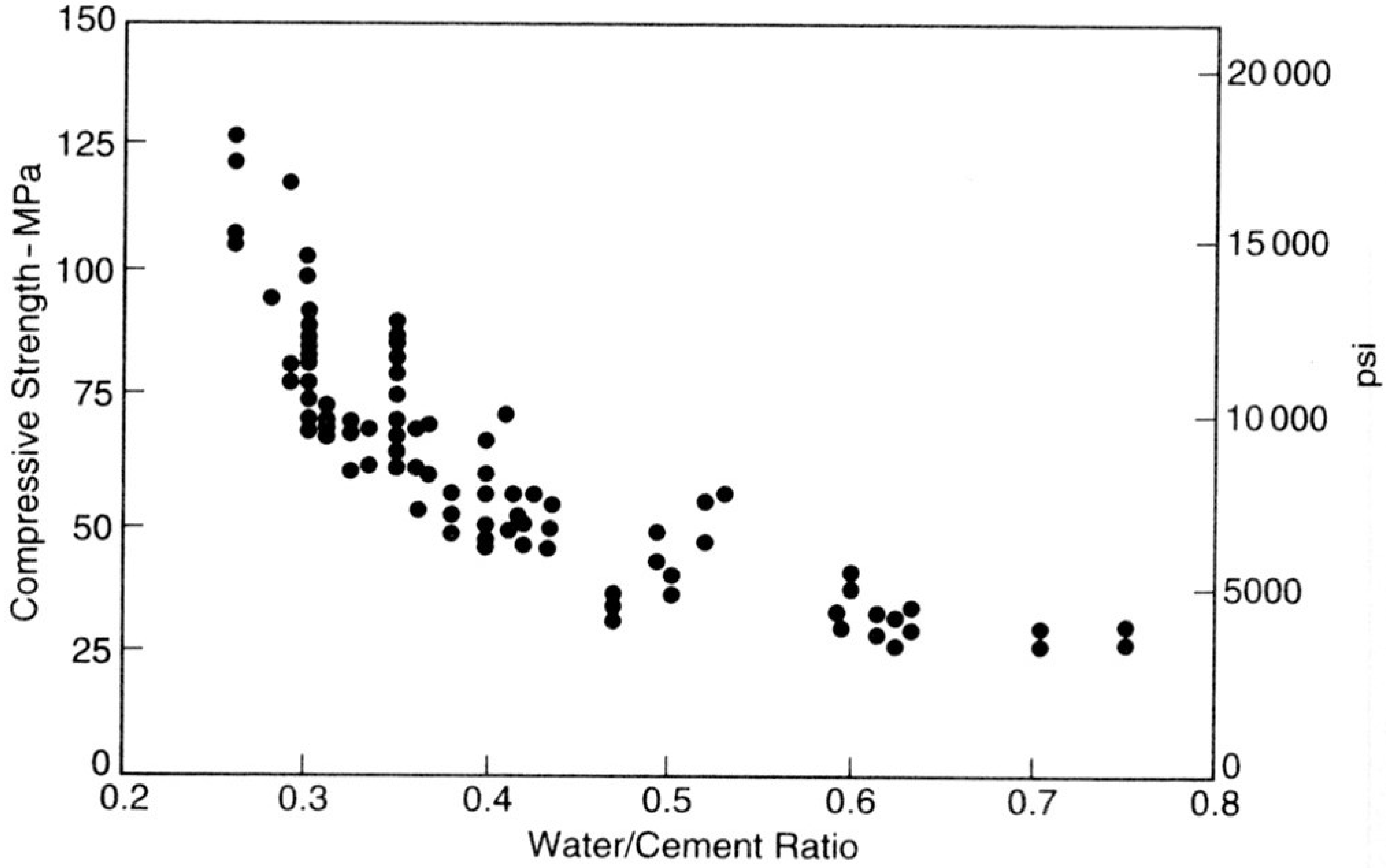
Aspects of HSC in the Fresh State

- Very high cement content, very low water content, high dosage of superplasticizer
- Batching and mixing require particular care
- Mixing

  ✓Use the mixer at less than its rated capacity (2/3 or ½ of the capacity) → thorough mixing

  ✓A longer mixing time (> 90 sec) than usual is required to ensure homogeneity

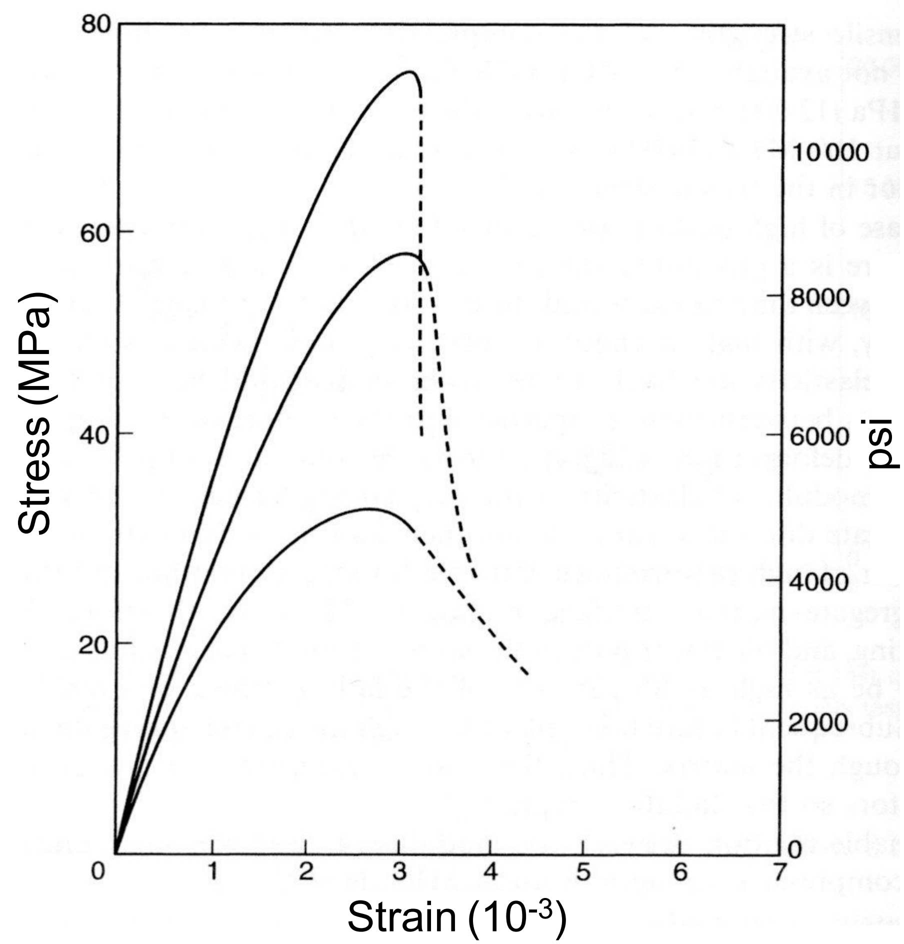
Aspects of Hardened HSC – Strength vs W/C Ratio



Aspects of Hardened HSC – Strength vs Brittleness

- The higher the strength the more brittle the concrete

- There are indications that, for compressive strengths above about 100 MPa, there is no further increase in the modulus of rupture or in the tensile strength



Aspects of Hardened HSC - Testing

- Due to a problem with respect to the capacity of the testing machine, the use of smaller specimens in preferable. Specifically, 100 by 200 mm cylinders or 100 mm cubes are satisfactory, given that, in high strength concrete, the maximum aggregate size is generally smaller than 12 mm. Such smaller test specimens give a measured strength about 5% higher than standard specimens.

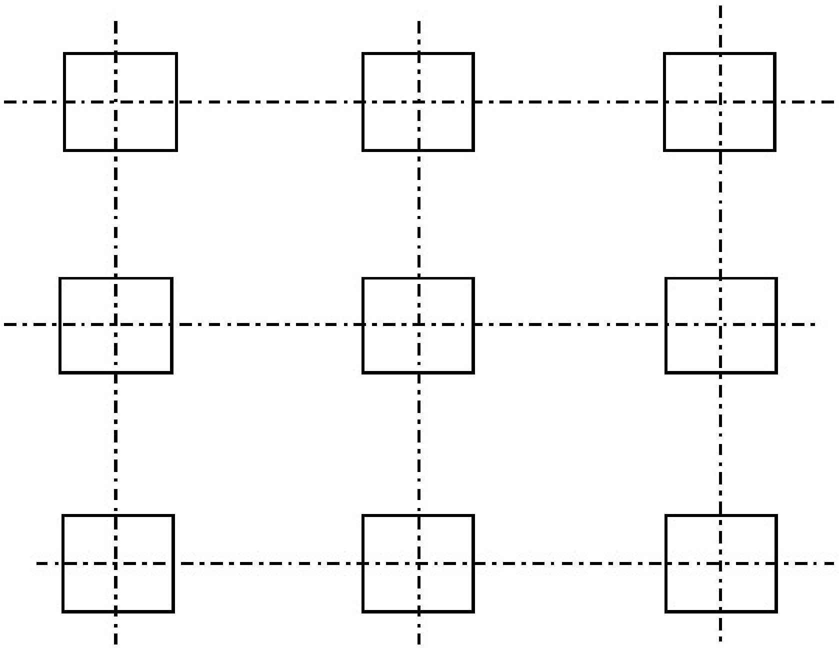
- The capping material used with test cylinders must not affect the recorded failure load on the specimen. For this reason, grinding of end surfaces is preferable.

Durability of HSC

- Extremely low permeability to chloride ions
- High resistant to alkali-silica reaction (ASR)
- The abrasion resistant of high-performance concrete is very good
- But high-performance concrete has a poor resistance to fire because the very low permeability of high performance does not allow the egress of steam formed from water in the hydrated cement paste

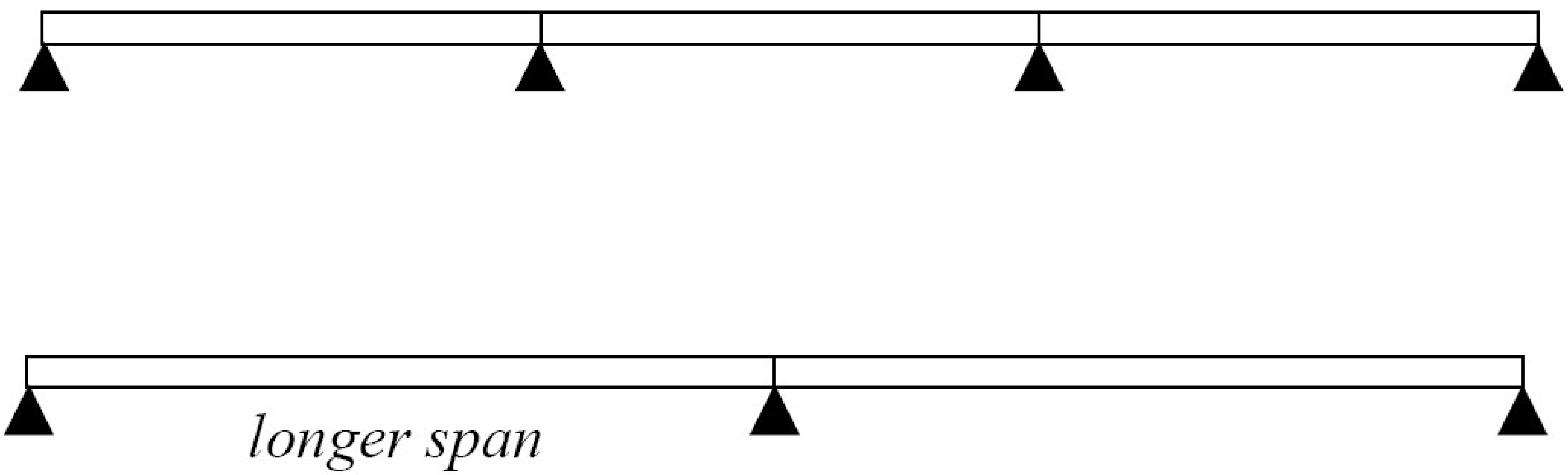
Advantages of HSC

- Column sections can be reduced in size or, for the same cross-section, the amount of steel reinforcement can be reduced
  - In tall buildings, there is an economic advantage because of an increased floor area for rental



Advantages of HSC

In bridges, the use of high strengths can reduce the number of beams.



Disadvantages of HSC

- Relatively low shear strength due to increased brittleness

- Increased creep and shrinkage due to a lower aggregate content

HSC Mix Design: Rules of Thumb

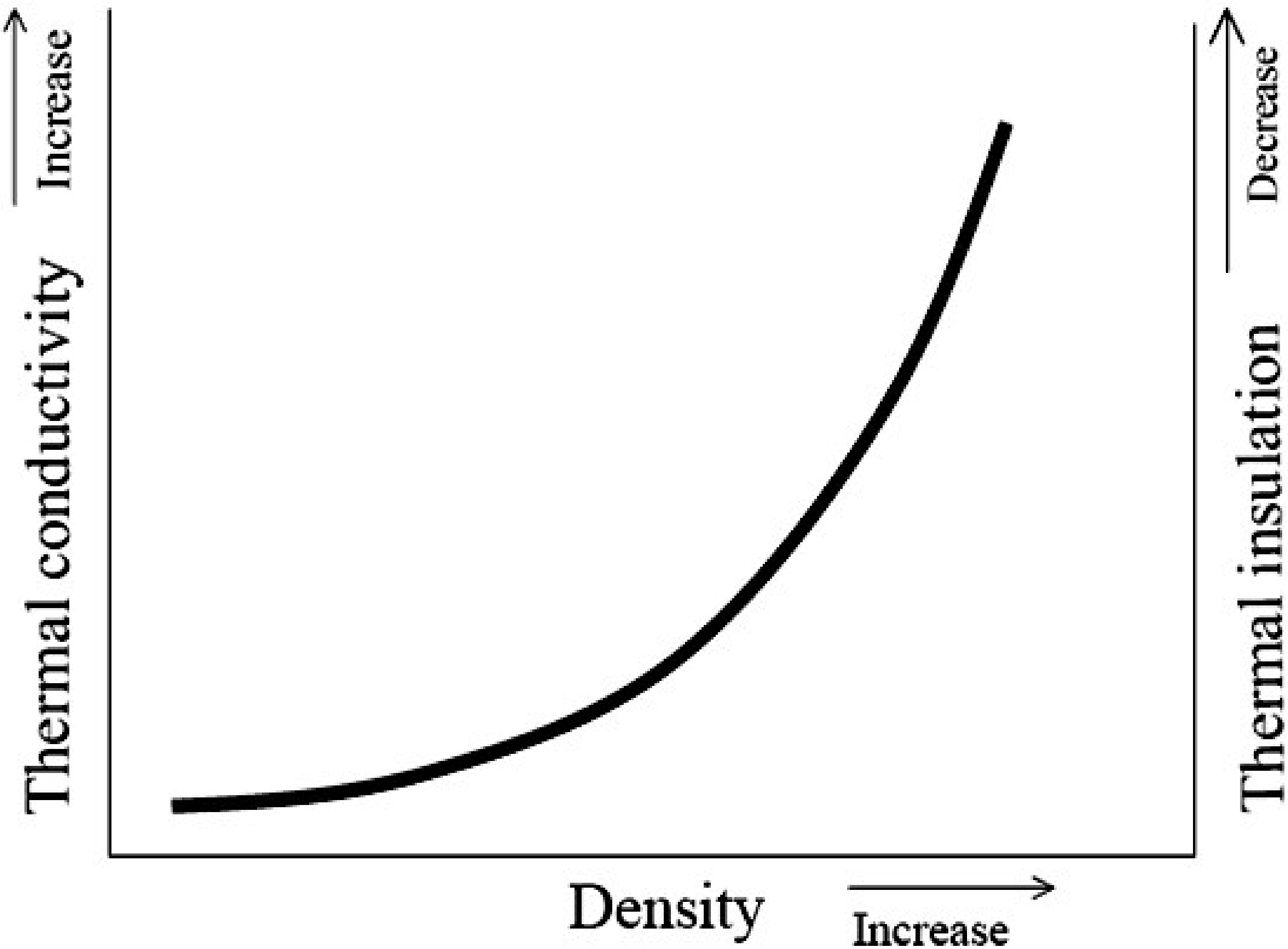
- Use lower W/CM Ratio
- Use higher cement content
- Use superplasticizer
- Use mineral admixtures (SF, FA, Slag)
- Use high strength coarse aggregate with smaller max size
- ***Expect higher cost***

Lightweight Concrete (LWC, 300 ~ 1850 kg/m3)

Advantages

- The reduction of self-weight, thus using smaller sections and the corresponding reduction in the size of foundations
- The formwork need withstand a lower pressure than would be the case with ordinary concrete
- The total weight of materials to be handled is reduced with a consequent increase in productivity
- LWC gives better thermal insulation than ordinary concrete

Density vs Thermal Conductivity



Classification of LWC

- ACI 213R-87 uses density to categorize concrete according to its application
- Density and strength are largely concomitant
- **Structural lightweight concrete**
  - Density: 1350 to 1900 kg/m^3
  - Compressive strength > 17 MPa
- **Moderate strength concrete**
  - Density: 800 to 1350 kg/m^3
  - Compressive strength: 7 to 17 MPa
- **Low Density Concrete**
  - Density: 300 to 800 kg/m^3
  - Non-structural purposes, mainly for thermal insulation

Ways to Achieve Light Weight

The density of concrete can be reduced by replacing some of the solid material in the mix by air voids. There are three possible locations of the air

- In the aggregate particles → ***lightweight aggregate concrete***
- In the cement paste → ***cellular concrete***
- Between the coarse aggregate particles, the fine aggregate being omitted → ***no-fines concrete***

Lightweight Aggregate Concrete

By using porous lightweight aggregates

- Natural lightweight aggregates
  - Diatomite, pumice, scoria, volcanic cinders, and tuff
- Manufactured lightweight aggregates
  - Expanded clay, shale, and slate
  - Vermiculite
  - Perlite
  - Expanded blastfurnaceslag
  - Clinker aggregate
  - Breeze

Strength of Lightweight Aggregate Concrete

For any particular aggregate, strength increases with an increase in density (cement content)

20 MPa → 260 to 330 kg/m^3 of cement

40 MPa → 420 to 500 kg/m^3 of cement

70 MPa → 630 kg/m^3 of cement

Cellular Concrete

- By introducing large voids within the concrete or mortar mass. This type of concrete is variously known as aerated, cellular, foamed or gas concrete

Void is introduced by …

- Aerating through chemical reaction: aluminum powder, hydrogen peroxide or calcium carbide as aerating agent

$Al + 2OH^- + 2\,H_2O \rightarrow Al(OH)_4^- + H_2\uparrow$

$2\,H_2O_2 \rightarrow 2\,H_2O + O_2\uparrow$

$CaC_2 + 2\,H_2O \rightarrow Ca(OH)_2 + C_2H_2\uparrow$ (acetylene)

- Blending pre-foamed foam: foaming agents used are detergents, resin soap etc.

No-Fines Concrete

- By omitting the fine aggregate from the mix so that a large number of interstitial voids is present. This concrete is know as *no-fines concrete*

- The coarse aggregate particles are cemented together by a layer of hardened cement paste (up to 1.3 mm) at their point of contact.

Aggregate Size for No-Fines Concrete

- Low density no-fines concrete is obtained with one-size aggregate
- The usual aggregate size is 10 to 20 mm
- 5% oversize and 10% undersize are allowed
- No particles should be smaller than 5 mm

Fresh Properties, Placing and Curing of No-Fines Concrete

- No workability tests for no-fines concrete
- Visual check to ensure even coating of all particles is adequate
- No-fines concrete must be placed very rapidly because
- the thin layer of cement paste can dry out
- No compaction during placing
- No-fines concrete does not segregate
- Moist curing is important because of the small thickness of cement paste involved

Hardened Properties of No-Fines Concrete

- Density

  With normal weight aggregate: 1600 to 2000 kg/m3

  With light weight aggregate: as low as 640 kg/m3

- Compressive strength

  Varies generally between 1.5 and 14 MPa

  Depends on its density

  Independent of W/C ratio

- Low shrinkage
- Low thermal expansion

Typical Properties of No-fines Concrete

Aggregate/cement ratio by volume	Water/cement ratio by mass	Density		28-day compressive strength	
		kg/m^3	lb/ft^3	MPa	psi
6	0.38	2020	126	14	2100
7	0.40	1970	123	12	1700
8	0.41	1940	121	10	1450
10	0.45	1870	117	7	1000

Application of No-Fines Concrete

- Insulation (sound, heat)
  - Load-bearing walls in domestic buildings
  - In-filling panels in framed structures
- Improved drainage
  - Pervious pavement
  - Pervious retaining wall